Atharva A. Patel

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Summary

AI/ML Engineer building production-grade Large Language Model (LLM) and Computer Vision systems. Experienced in developing end-to-end Retrieval-Augmented Generation (RAG) pipelines, AI agent workflows, and real-time object detection solutions with a focus on performance, accuracy, and scalability. Strong foundation in Python, PyTorch, LangChain, OpenCV, and modern AI technologies.

Education

- **International Institute of Information Technology** — Pune, India
  Bachelor of Engineering - Information Technology; GPA: 7.27 — *July 2023 - June 2027*
  Courses: *Operating Systems, Data Structures, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases*

Skills Summary

- **Languages**: Pythonproficient, C++, JavaScript, SQL, Bash, JAVA
- **AI/LLM Concepts**: Pretraining, Transformers, Attention, RAG, Evals, Hallucination Analysis, Agent Workflows
- **Frameworks**: Scikit, NLTK, SpaCy, TensorFlow, Keras, Django, Flask, NodeJS, LAMP
- **Tools**: Kubernetes, Docker, GIT, PostgreSQL, MySQL, SQLite
- **Platforms**: Linux, Web, Windows, Arduino, Raspberry, AWS, GCP, Alibaba Cloud, IBM Cloud
- **Computer vision**: CNN, YOLO, OpenCV, Image Processing, NLP: Transformers, BERT, GPT, SpaCy, NLTK
- **Soft Skills**: Leadership, Event Management, Writing, Public Speaking, Time Management

Experience

- **AI/ML Student Researcher — I2IT Pune** — In-Campus
  Student Developer (Full-time) — *May 2025 - Sep 2025*
  - **AI Agent Research**: Conducted independent research on AI agents, machine learning models, and intelligent automation techniques.
  - **Autonomous AI System**: Developed autonomous AI applications and real-time computer vision solutions using modern deep learning frameworks.
- **DataCamp Inc.** — Remote
  Instructor (Part-time, Contractual) — *Dec 2025*
  - **Project Course - Find Movie Similarity from Plot Summaries**: Created project based course using Unsupervised learning and natural language processing.
  - **Tutorial - Introduction to Reinforcement Learning**: Created tutorial for Q-learning RL algorithm and concepts.
  - **Impact**: Course has been taken by 250+ students so far with 4.65 average rating.

Projects

- **Autonomous Research Agent with RAG System — Python, LangChain, OpenAI API, FAISS, Streamlit)**: - Engineered AI assistant processing 15+ data sources, reducing research time by 80percent - Implemented RAG with FAISS achieving 92% accuracy, reducing hallucinations by 75% - Deployed Streamlit web interface handling 500+ queries with ¡2 second response time
- **Sentinel — Reliability Guardrail Layer for Multi-Agent Systems (Multi-Agent Systems, LLM Orchestration, Reliability Engineering)**: Built a supervisor framework enforcing bounded loops, schema-validated tool calls, and cost/token budgets across specialized executor agents, with a critic/verifier gate and event-sourced deterministic replay; built a golden-scenario eval harness measuring [75-90]% task success rate and [90-99]% tool-call validity. Tech: Python, LangGraph, Pydantic, FastAPI, Redis, Postgres, Docker/Kubernetes. (2026)
- **InfernoServe — Continuous-Batching LLM Inference Engine (LLM Systems, GPU Programming, Distributed Serving)**: Built a from-scratch LLM inference engine with paged KV-cache, continuous batching, speculative decoding, and a goodput-aware scheduler, reaching [1000-1500] tokens/sec within [65-75]% of vLLM. Tech: Python, C++/CUDA, PyTorch, Triton, FastAPI, Docker, Kubernetes. (2026)
- **GradMesh — Distributed Data-Parallel Training Framework (Distributed Systems, GPU Performance, Fault Tolerance)**: Built a custom ring all-reduce and gradient compression pipeline overlapping communication with backprop across multiple GPUs, with coordinator-managed elastic membership and checkpoint-based fault recovery; benchmarked [85]% strong-scaling efficiency vs ideal linear on 2 GPUs. Tech: Python, PyTorch, NCCL/Gloo, CUDA streams, Kubernetes, Prometheus. (2026)
- **MiniGPT: Transformer Language Model from Scratch(PyTorch, FastAPI, Docker)**: Built and publicly documented a 22M-parameter, six-block GPT-style model with causal multi-head attention, pre-layer normalization, AdamW, cosine scheduling, and gradient clipping; added WebSocket token streaming and reproducible setup instructions. (2026)

Achievements

- **Kaggle Contributor** – Top 15% in ML competitions with 2500+ participants.
- **Hackathon Finalist** – Developed AI-powered tool in a 24-hour coding marathon.
- **Open Source Contributor** – Active in AI/ML repositories on GitHub.

Extra Curriculars

Athletics & Marathon Runner — Sports Volunteer — Esports (CSGO)